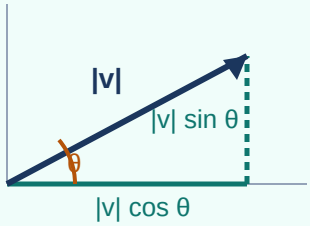
Vectors Practice

Challenge set — component form, length and direction, bearings, and velocity — from the Wednesday, November 4 session.

Name: Ryan Naeimi **Time:** about 45 minutes · 12 problems **Suggested due:** Wed, Nov 11

KEY IDEAS — THE TRIG DEFINITION

- A vector with length $|v|$ and direction angle θ (measured from the positive x-axis) has components $\boldsymbol{v} = \langle\, |\boldsymbol{v}| \cos\theta,\ |\boldsymbol{v}| \sin\theta \,\rangle$. Cosine always goes with x, sine with y.
- Going backwards: **length $|\boldsymbol{v}| = \sqrt{a^2 + b^2}$** and **$\tan\theta = b / a$** for $v = \langle a, b \rangle$. Check the quadrant before choosing the angle.
- **Unit vector** in the direction of v: $\boldsymbol{u} = \boldsymbol{v} / |\boldsymbol{v}|$. It has length 1 and keeps the direction.



A vector is its two components put together: run across, then rise up.

- **Bearings** are measured clockwise from north — so the trig swaps: **east = $|\boldsymbol{v}|$ sin(bearing)**, **north = $|\boldsymbol{v}|$ cos(bearing)**.
- **Ground velocity = air velocity + wind velocity**, added component by component.
- **Heading is not track.** Heading is where the nose points; track is where you actually go. Aim into a sideways wind or current so the track comes out right (Problems 10–11).

Show your work. Give exact values where they are clean, otherwise round to 2 decimal places (angles to 1 decimal place).

A — Components using trig

about 7 min

1 Find the component form of the vector with length 42 and direction angle 213°. Predict the signs from the quadrant first, then compute.

2 **Work backwards.** A vector has length 30 and horizontal component −18, and it points into the third quadrant. Find (a) its vertical component and (b) its direction angle.

B — Length, direction, and unit vectors

about 8 min

3 Let $\mathbf{a} = \langle 3, -7 \rangle$ and $\mathbf{b} = \langle -5, 2 \rangle$. For $\mathbf{w} = 2\mathbf{a} - 3\mathbf{b}$, find (a) component form, (b) length, (c) direction angle, and (d) its unit vector.

4 Let $\mathbf{u} = \langle -6, 8 \rangle$. Find (a) the unit vector in the direction *opposite* to $\mathbf{u}$, and (b) the vector of length 35 in that opposite direction.

C — Bearings

about 7 min

5 A drone flies 12 km on a bearing of 205°. Find its east and north components, and state in words which directions those signs mean it actually travelled.

6 A displacement ends 18 km east and 11 km south of its start. Find (a) its length, (b) its bearing, and (c) the unit vector in its direction, using (east, north) components.

D — Adding vectors and forces

about 8 min

7 Vector **a** has length 26 and direction angle 68°. Vector **b** has length 19 and direction angle 241°. Find the resultant **a** + **b**: its component form, its length, and its direction angle. Before computing, predict: will the resultant be longer or shorter than both **a** and **b**? Why?

8 **Equilibrium.** Force $\mathbf{F}_1$ is 80 N at a direction angle of 35°, and force $\mathbf{F}_2$ is 60 N at a direction angle of 140°. (a) Find the resultant force: components, magnitude, and direction angle. (b) A third force holds the object still (equilibrium). Find that force's components, magnitude, and direction angle.

READING A BEARING

A bearing starts at **north** and turns **clockwise**. Bearing 000° is north, 090° is east, 180° is south, and 270° is west. Problem 5 uses a bearing of 205°: the direction has turned past south, so both components come out negative — west and south. When you convert a bearing, east gets **sin** and north gets **cos**, the opposite order from a standard direction angle.

E — Word problems: velocity and navigation

about 15 min

THE ONE RULE FOR THIS SECTION

Ground velocity = air velocity + wind velocity. Write both velocities in (east, north) components, add them, then find the speed and bearing of the result. In Problems 10 and 11 the *track* is the one you are given — work backwards to the heading that produces it, so the sideways push cancels instead of adding.

9 **Plane and wind.** A plane heads on a bearing of 060° at an airspeed of 260 mph. The wind blows at 50 mph from the northwest (so the wind itself points southeast, on a bearing of 135°). Find the plane's ground velocity in component form (east, north), its ground speed, and its bearing. Is the ground bearing larger or smaller than the heading? Explain in one line.

10 **Holding a course.** A plane with airspeed 250 mph needs its ground track to be due east, but a wind blows from the northwest at 40 mph (toward the southeast). What bearing should the pilot head, and what will the ground speed be?

11 **River crossing, aiming upstream.** A boat that travels at 12 km/h in still water must land *directly across* a river whose current flows east at 5 km/h. (a) On what bearing should the boat head, and what is its ground speed straight across? (b) The river is 1.30 km wide — how long does the crossing take?

12 **Two legs of a voyage — and the way home.** A ship sails 45 km on a bearing of 038°, then 28 km on a bearing of 172°. (a) How far is the ship from its starting point, and on what bearing from the start? (b) The ship now sails straight home. What bearing should it follow, and how far is that leg? *Hint: convert each leg to east/north components first, then add.*

Stuck? Every problem on this sheet starts the same way: turn each vector into components first. Magnitude and angle → use cos and sin. Bearing → use sin for east, cos for north. Two velocities → add the components — unless the track is given, in which case subtract the push and solve for the heading. The answer key on the next pages works every problem in full — try first, then check step by step.

Worked Answer Key

Work each problem first, then compare step by step — the step where your work first differs is the one to practise.

Problems 1–6 Exact values first, decimals after

1 213° is in the **third quadrant**, so predict both components negative. x = 42 cos 213° = 42(−0.8387) = −35.22; y = 42 sin 213° = 42(−0.5446) = −22.87. Both signs match the prediction.✓

> **Answer:** ⟨−35.22, −22.87⟩

2 (a) Components and length obey $a^2 + b^2 = |v|^2$: $b^2 = 30^2 - (-18)^2 = 900 - 324 = 576$, so b = ±24. Third quadrant means the vertical component is also negative: b = −24. (b) Reference angle = $\tan^{-1}(24/18) = 53.1°$; third quadrant, so θ = 180° + 53.1° = 233.1°.

> **Answer:** vector ⟨−18, −24⟩; direction 233.1°

3 (a) $2a = \langle 6, -14\rangle$ and $3b = \langle -15, 6\rangle$, so $w = \langle 6 - (-15), -14 - 6\rangle = \langle 21, -20\rangle$. (b) $|w| = \sqrt{(21^2 + 20^2)} = \sqrt{841} = 29$. (c) Fourth quadrant: θ = 360° − $\tan^{-1}(20/21)$ = 360° − 43.6° = 316.4°. (d) Unit vector = ⟨21/29, −20/29⟩ ≈ ⟨0.72, −0.69⟩.

> **Answer:** ⟨21, −20⟩; length 29; direction 316.4°; unit vector ⟨21/29, −20/29⟩

4 $|u| = \sqrt{((-6)^2 + 8^2)} = \sqrt{100} = 10$. Unit vector along *u* is ⟨−0.60, 0.80⟩. (a) **Opposite** flips both signs: ⟨0.60, −0.80⟩. (b) Scale by 35: 35 × ⟨0.60, −0.80⟩ = ⟨21, −28⟩. Check: $\sqrt{(21^2 + 28^2)} = 35$.✓

> **Answer:** (a) ⟨0.60, −0.80⟩ (b) ⟨21, −28⟩

5 **Bearing trig swaps:** east = 12 sin 205° = 12(−0.4226) = −5.07 km; north = 12 cos 205° = 12(−0.9063) = −10.88 km. Negative east means west, negative north means south.

> **Answer:** east −5.07 km and north −10.88 km — the drone travelled 5.07 km west and 10.88 km south

6 In (east, north) components the displacement is ⟨18, −11⟩ km. (a) **Length:** $\sqrt{(18^2 + 11^2)} = \sqrt{445} \approx 21.10$ km. (b) **Bearing:** southeast quadrant, so bearing = 180° − $\tan^{-1}(18/11)$ = 180° − 58.6° = 121.4°. (c) **Unit vector:** ⟨18/21.10, −11/21.10⟩ ≈ ⟨0.85, −0.52⟩.

> **Answer:** length about 21.10 km; bearing 121.4°; unit vector ⟨0.85, −0.52⟩

7 **a** = ⟨26 cos 68°, 26 sin 68°⟩ = ⟨9.74, 24.11⟩. **b** = ⟨19 cos 241°, 19 sin 241°⟩ = ⟨−9.21, −16.62⟩. Add: ⟨9.74 − 9.21, 24.11 − 16.62⟩ = ⟨0.53, 7.49⟩. Length = $\sqrt{(0.53^2 + 7.49^2)} \approx 7.51$. Direction = $\tan^{-1}(7.49/0.53) = 86.0°$ (first quadrant). The x-components nearly cancel — that is why the resultant is *shorter than both* vectors that made it.

> **Answer:** ⟨0.53, 7.49⟩; length about 7.51; direction 86.0°

8 $\mathbf{F}_1$ = ⟨80 cos 35°, 80 sin 35°⟩ = ⟨65.53, 45.89⟩ N. $\mathbf{F}_2$ = ⟨60 cos 140°, 60 sin 140°⟩ = ⟨−45.96, 38.57⟩ N. (a) Resultant = ⟨19.57, 84.45⟩ N; magnitude = $\sqrt{(19.57^2 + 84.45^2)} \approx 86.69$ N; direction = $\tan^{-1}(84.45/19.57) = 77.0°$. (b) Equilibrium force is the exact negative: ⟨−19.57, −84.45⟩ N, same magnitude 86.69 N, direction 77.0° + 180° = 257.0°.

> **Answer:** (a) ⟨19.57, 84.45⟩ N, about 86.69 N at 77.0° (b) ⟨−19.57, −84.45⟩ N, about 86.69 N at 257.0°

9 Air velocity (heading bearing 060°): east = 260 sin 60° = 225.17, north = 260 cos 60° = 130.00. Wind toward the southeast (bearing 135°): east = 50 sin 135° = 35.36, north = 50 cos 135° = −35.36. Ground = air + wind = ⟨260.52, 94.64⟩ mph. Ground speed = $\sqrt{(260.52^2 + 94.64^2)} \approx 277.18$ mph. Bearing = $\tan^{-1}(260.52/94.64) = 070.0°$. The bearing is *larger* than the 060° heading: the wind's southward push rotates the track clockwise, toward the east.

> **Answer:** ground velocity ⟨260.52, 94.64⟩ mph; ground speed about 277.18 mph; bearing 070.0°

10 Wind toward the southeast = ⟨40 sin 135°, 40 cos 135°⟩ = ⟨28.28, −28.28⟩ mph. For a due-east track, the air velocity's north component must cancel the wind's −28.28: air north = +28.28. Then air east = $\sqrt{(250^2 - 28.28^2)} = 248.39$. Heading bearing = $\tan^{-1}(248.39/28.28) = 83.5°$ — aim north of east. Ground speed = air east + wind east = 248.39 + 28.28 = 276.68 mph due east.

> **Answer:** head on bearing 083.5°; ground speed about 276.68 mph due east

11 (a) To land directly across, the boat's westward component must cancel the 5 km/h eastward current: $\sin\theta = 5/12$, so $\theta = 24.6°$ west of north, a bearing of 360° − 24.6° = 335.4°. Ground speed straight across = $\sqrt{(12^2 - 5^2)} = \sqrt{119} \approx 10.91$ km/h due north. (b) Time = 1.30 / 10.91 = 0.119 h ≈ 7.15 minutes.

> **Answer:** (a) head on bearing 335.4°, ground speed about 10.91 km/h (b) about 7.15 minutes

12 Leg 1: east = 45 sin 038° = 27.70 km, north = 45 cos 038° = 35.46 km. Leg 2: east = 28 sin 172° = 3.90 km, north = 28 cos 172° = −27.73 km. Totals: east = 31.60 km, north = 7.73 km. (a) Distance = $\sqrt{(31.60^2 + 7.73^2)} \approx 32.53$ km. Bearing = $\tan^{-1}(31.60/7.73) = 76.2°$. (b) Home is the opposite direction: bearing 76.2° + 180° = 256.2°, and the leg home is the same 32.53 km.

> **Answer:** (a) 32.53 km from the start, on a bearing of 076.2° (b) sail home on bearing 256.2°, for 32.53 km

Before the next session: redo any problem where your first differing step was the setup — choosing cos vs. sin, sin vs. cos for a bearing, which quadrant the angle lands in, or whether the wind adds to the track or has to be cancelled out of it. Those choices decide almost every problem on this sheet.